

Assembly Production Line Design.

Let's assume based on demand and supply, the Marketing team asked to produce 10,000 units per year.

Hourly rate of production (r), needed to satisfy this demand with the following factors:

- Factory operates for 8 hours shift per day and 250 days a year.

$$r = \frac{\text{Sales}}{\text{time}} = \frac{10,000 \frac{\text{units}}{\text{year}}}{250 \frac{\text{days}}{\text{yr}} \cdot 8 \frac{\text{hours}}{\text{day}}} = \frac{10000}{250 \cdot 8} = 5 \text{ units/hour}$$

Thus, takt time t, for the process is:

$$t = \frac{1}{r} = \frac{1}{5} \text{ hour} = 12 \text{ minutes}$$

Line Efficiency let's assume the line is running at 100% efficiency. (Due to factors like rework or downtime, we might need a shorter takt time to compensate.)

Each assembly line workstation has 12 minutes of work, and the total assembly time is estimated per printer to be 4.8 hours, then we need about $4.8 \cdot 60 / 12 = 24$ workstations.

Prusa Printer								by Factor 10
#	Name	Parts (inc Fasteners)	Parts (no fasteners)	Adjustments	Time	Necessary parts	Efficiency	Expected Time (min)
A1	Frame Assembly	45	11	1	78.97	2	8%	13.2
A2	Y Axis Assembly	73	21	2	236.41	9	11%	39.4
A3	X Axis Assembly	21	14	4	130.23	6	14%	21.7
A4	Z Axis Assembly	32	10	6	141.18	4	8%	23.5
A5	Extruder Assembly	78	31	31	475.22	15	9%	79.2
A6	LCD Assembly	15	5	4	62.20	3	14%	10.4
A7	Heatbed Assembly	35	14	2	305.00	3	3%	50.8
A8	PSU Assembly	11	7	1	91.05	3	10%	15.2
A9	Electronics Assembly	37	9	22	198.25	3	5%	33.0
A10	Spool Holder + Assembly	5	5	0	20.75	1	14%	3.5
	Prusa MK3 Printer	352	127	73	1739.26	49	8%	4.8 hrs
					29.0	minutes		

By considering a factor of 10, Estimated assembly time for A5 extruder is = 80 min

Total assembly time needed for A-5 Extruder Assembly: 80 minutes (From DFA analysis and estimation)

So Total workcells needed: $\frac{80 \text{ minutes (A5 ASSEMBLY TIME)}}{12 \text{ min(Takt time)}} = 6.67 \text{ workcells}$

i.e. = 7 workcells for A5 Extruder assembly

Below are the estimated division of works between workcells.

Assembly Process for A5 Extruder Assembly

Workcell Design and Assembly Steps

To ensure the A5 Extruder Assembly is completed within the takt time of 12 minutes, the assembly process will be divided into 7 workcells. Each workcell will perform a series of logically sequenced tasks, ensuring that all work is completed efficiently and within the time constraints.

Workcell 1: Initial Assembly (Extruder Body , FS Lever and Steel ball Assembly)

- **Tasks:**
 - Assemble the Extruder Body (5.1-5.5) [0.58 min]
 - Assemble and install FS Lever (5.5-5.6) [2.68 min]
 - Assemble Steel Ball (5.7) [2.21 min]
- **Total Time: 5.47 minutes**
- **Workers: 1**

Workcell 2: Intermediate Assembly (Bondtech Gear Assembly)

- **Tasks:**
 - Assemble Bondtech Gear (5.9-5.14) [11.06 min]
- **Total Time: 11.06 minutes (Workcell will focus on Bondtech Gear)**
- **Workers: 1**

Workcell 3: X Carriage Assembly

- **Tasks:**
 - Assemble X Carriage (5.15-5.19) [18.27 min]
 - Includes tasks like routing motor cables, aligning X carriage, tightening screws, attaching IR sensor and hot end fan.
- **Total Time: 18.27 minutes (split over two persons averaging 9.14min per person)**
- **Workers: 2**

Workcell 4: Extruder Idler Assembly

- **Tasks:**

- Assemble Extruder Idler (5.32-5.36) [6.07 min]
- **Total Time: 6.07 minutes**
- **Workers: 1**

Workcell 5: FS Cover and Print Fan Support Assembly

- **Tasks:**
 - Assemble FS Cover
 - Assemble Print Fan Support (5.40-5.41)
 - Assemble Fan Shroud, PINDA Sensor (5.42-5.47)
- **Total Time: 9.9 minutes (1 worker)**
- **Workers: 1**

Workcell 6: Cable alignments, zip tie, and X-axis belt installation

- **Tasks:**
 - Cable alignments, zip tie, and X-axis belt installation
- **Total Time: 28.43 minutes (split into three people working in parallel)**
- **Workers: 3**

Workcell 7: Final Inspection and Testing

- **Tasks:**
 - Quality inspection and testing for alignment, tensioning, and ensuring all components are secure.
 - Ensure all screws are tight, no components are missing, and the assembly is functional.
- **Total Time: 12 minutes**
- **Workers: 1**

Additional Roles:

- Forklift Operator: 1 (for moving assemblies between workcells)
- Rework Expert: 1 (for handling any rework tasks)
- Inspection Person: 1 (for final inspection and quality control)

Total Number of Workers: 10 assembly workers + 3 additional roles = 13 workers

Preliminary assembly line layout floorplan